

ACTIVITY 04

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Question 1 – Empirical Cumulative Distribution Function (ECDF)

Scenario

A teacher recorded the marks of students in a Data Science quiz. The teacher wants to determine the cumulative proportion of students scoring at or below a given mark.

Dataset

Student	Marks
S1	45
S2	52
S3	58
S4	60
S5	65
S6	68
S7	72
S8	75
S9	80
S10	90

Question

Construct the **Empirical Cumulative Distribution Function (ECDF)** for the above dataset. Calculate the cumulative probability for each observation and interpret the ECDF plot.

Code

```
# Student Marks Dataset
```

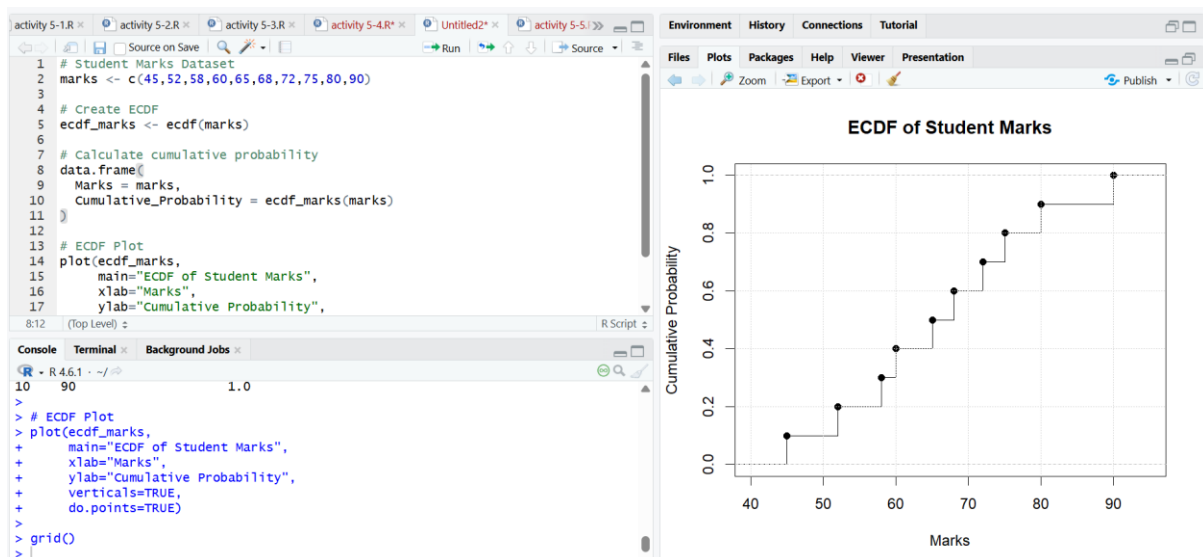
```
marks <- c(45,52,58,60,65,68,72,75,80,90)
```

```
# Create ECDF
```

```
ecdf_marks <- ecdf(marks)
```

```
# Display cumulative probability
```

```
data.frame(
  Marks = marks,
  Cumulative_Probability = ecdf_marks(marks)
)
# ECDF Plot
plot(ecdf_marks,
     main="ECDF of Student Marks",
     xlab="Marks",
     ylab="Cumulative Probability",
     verticals=TRUE)
grid()
```



Question 2 – Empirical Cumulative Distribution Function (ECDF)

Scenario

A fitness trainer recorded the number of daily steps completed by ten participants during a health challenge.

Dataset

Participant	Daily Steps
P1	4200

Participant	Daily Steps
P2	5000
P3	5600
P4	6100
P5	6800
P6	7200
P7	8100
P8	9200
P9	10400
P10	12000

Question

Using the dataset, create an **ECDF plot**. Determine the proportion of participants who walked **8,000 steps or fewer** and explain the significance of the ECDF.

Code

```
# Daily Steps Dataset

steps <- c(4200,5000,5600,6100,6800,
          7200,8100,9200,10400,12000)

# Create ECDF

ecdf_steps <- ecdf(steps)

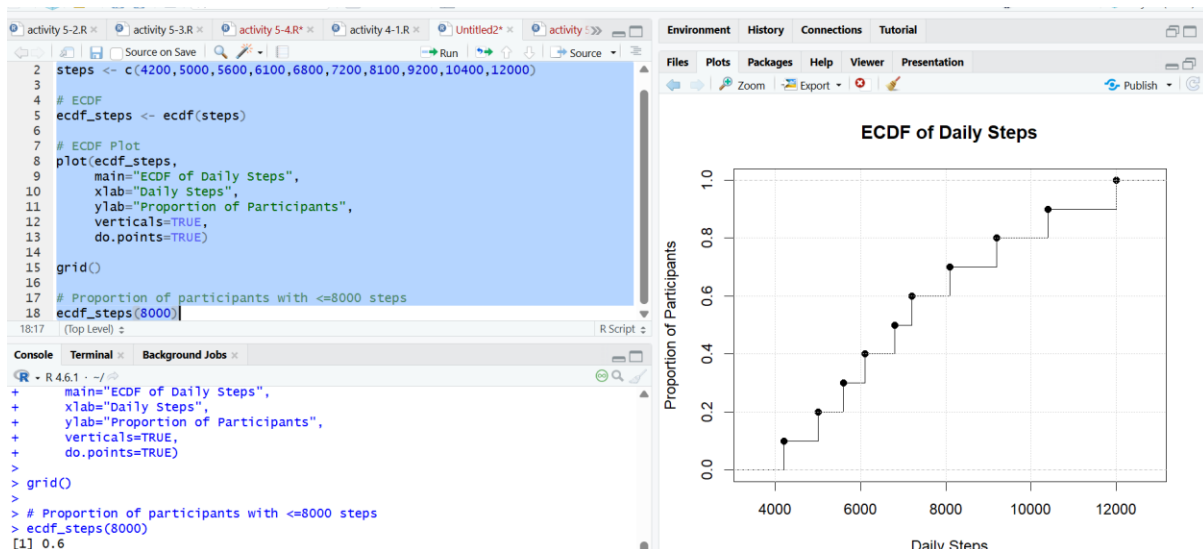
# ECDF Plot

plot(ecdf_steps,
     main="ECDF of Daily Steps",
     xlab="Daily Steps",
     ylab="Cumulative Proportion",
     verticals=TRUE)

grid()

# Proportion of participants with steps <= 8000

ecdf_steps(8000)
```



Question 3 – Highly Skewed Distribution (Histogram and Density Plot)

Scenario

An online shopping company recorded the purchase amount (₹) made by customers during a promotional sale. Most customers purchased inexpensive items, while a few made very expensive purchases.

Dataset

Customer	Purchase Amount (₹)
C1	250
C2	300
C3	320
C4	350
C5	380
C6	450
C7	520
C8	650
C9	1800
C10	5200

Question

Create a **Histogram** and a **Density Plot** for the purchase amounts. Determine whether the distribution is positively skewed or negatively skewed, and justify your answer.

Code

```
# Purchase Amount Dataset
```

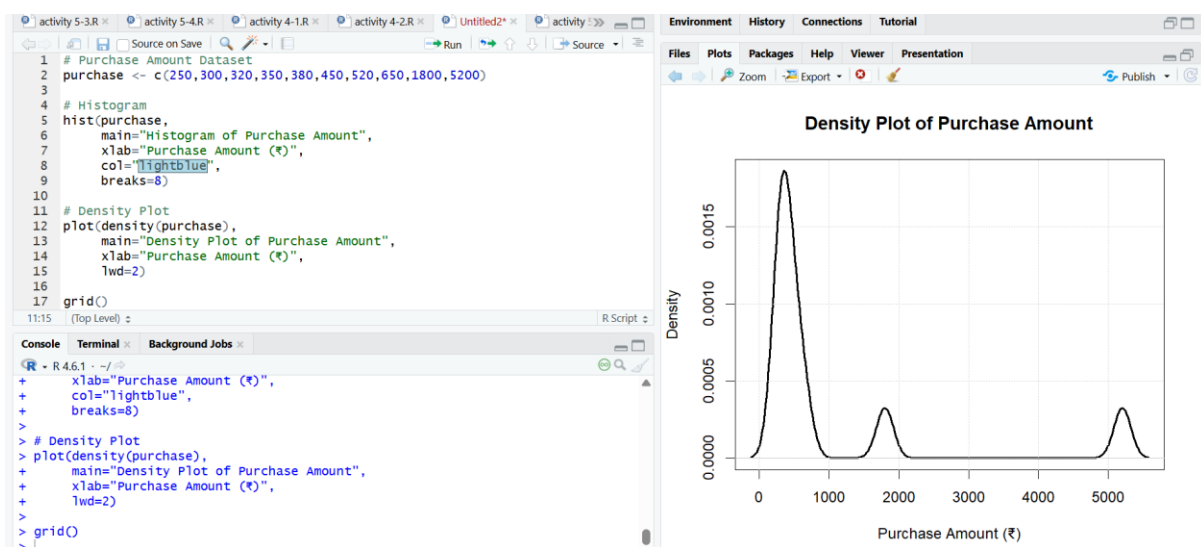
```
purchase <- c(250,300,320,350,380,  
              450,520,650,1800,5200)
```

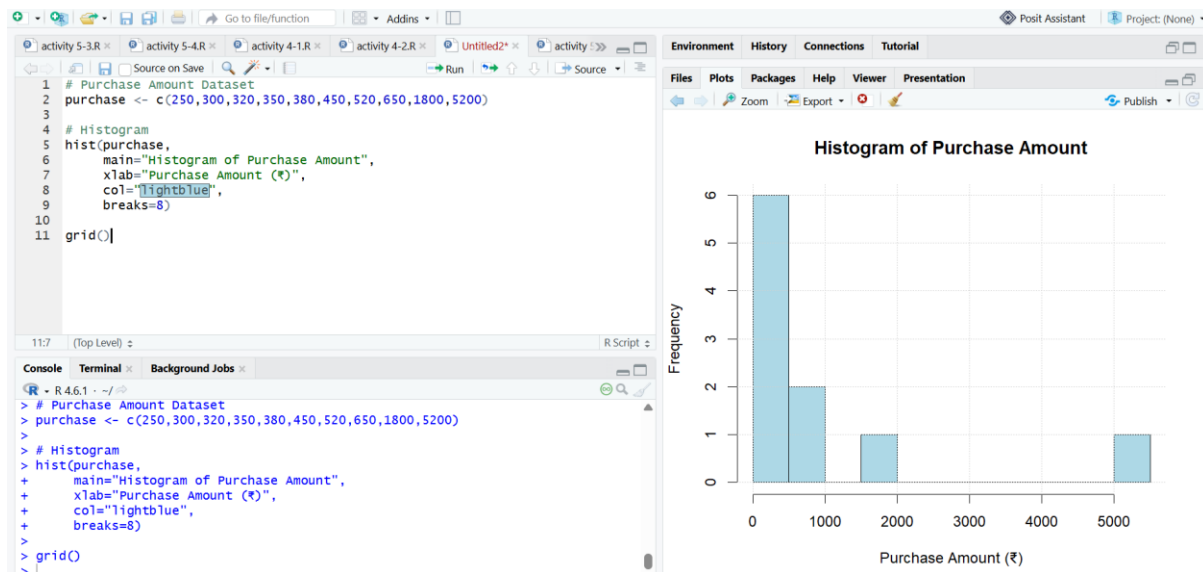
```
# Histogram
```

```
hist(purchase,  
     main="Histogram of Purchase Amount",  
     xlab="Purchase Amount (₹)",  
     breaks=8)
```

```
# Density Plot
```

```
d <- density(purchase)  
  
plot(d,  
     main="Density Plot of Purchase Amount",  
     xlab="Purchase Amount (₹)",  
     lwd=2)  
  
grid()
```





Question 4 – Quantile-Quantile (Q-Q) Plot

Scenario

A hospital administrator collected the waiting times (in minutes) for patients in the outpatient department and wants to verify whether the data follows a normal distribution.

Dataset

Patient	Waiting Time (minutes)
P1	12
P2	15
P3	18
P4	20
P5	22
P6	24
P7	25
P8	28
P9	30
P10	32

Question

Construct a **Normal Q-Q Plot** using the waiting time data. Based on the plot, determine whether the waiting times approximately follow a normal distribution.

Code

```
# Waiting Time Dataset

waiting_time <- c(12,15,18,20,22,
                 24,25,28,30,32)

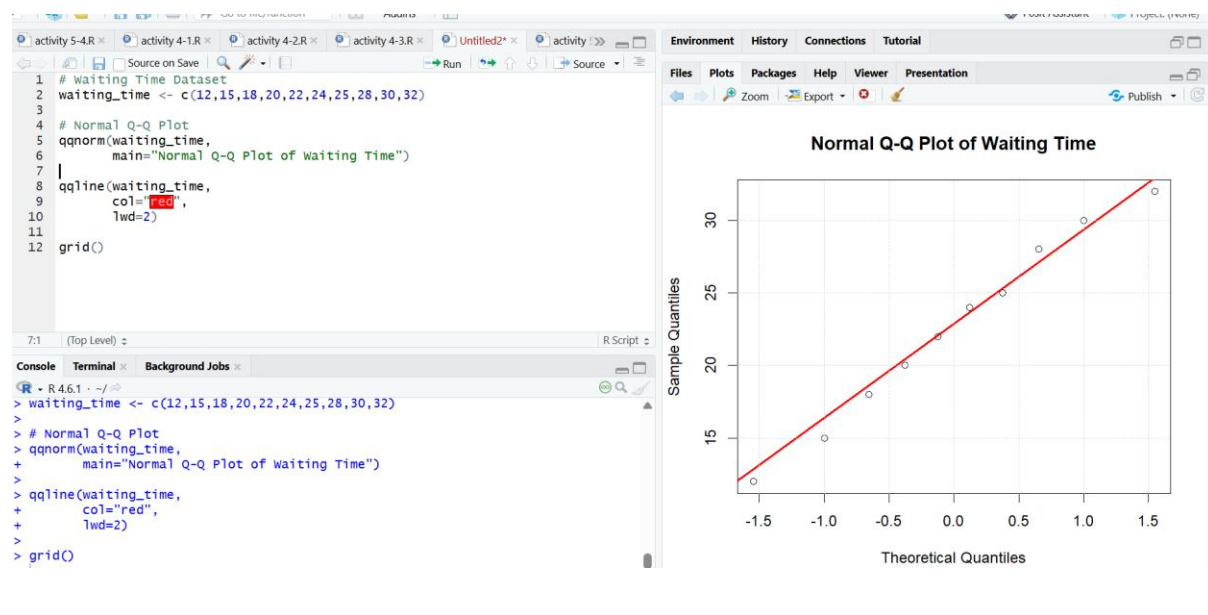
# Normal Q-Q Plot

qqnorm(waiting_time,
        main="Normal Q-Q Plot of Waiting Time")

# Reference Line

qqline(waiting_time,
        col="red",
        lwd=2)

grid()
```



Question 5 – Highly Skewed Distribution and Q-Q Plot

Scenario

A bank analyzed the monthly income (₹) of ten customers applying for home loans. A few customers have exceptionally high incomes compared to the majority.

Dataset

Customer	Monthly Income (₹)
C1	18000
C2	22000
C3	24000
C4	26000
C5	28000
C6	30000
C7	34000
C8	42000
C9	85000
C10	200000

Question

1. Plot a **Histogram** of the monthly income data.
2. Generate a **Normal Q-Q Plot**.
3. Determine whether the income data is normally distributed or highly skewed.
4. Explain how the Q-Q plot supports your conclusion.

Code

```
# Monthly Income Dataset

income <- c(18000,22000,24000,26000,28000,
           30000,34000,42000,85000,200000)

# Histogram

hist(income,
     main="Histogram of Monthly Income",
     xlab="Income (₹)",
     breaks=8)

# Normal Q-Q Plot

qqnorm(income,
```



```
main="Normal Q-Q Plot of Monthly Income")
```

```
# Reference Line
```

```
qqline(income,
```

```
col="red",
```

```
lwd=2)
```

```
grid()
```

